Younes Boutiyarzist

R&D Engineer / PhD Candidate in Autonomous Systems

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R&D engineer and PhD candidate specializing in **sensor fusion**, and **AI-based perception** for real-world aerospace systems. My work focuses on **state estimation**, **multi-sensor fusion**, **Kalman filter related methods**, and robust algorithm design under uncertainty, with applications to drone and helicopter platforms. I am particularly interested in **navigation**, **autonomous flight**, and the validation of algorithms through **simulation**, **ground tests**, and **real-world flight data**.

Experience

R&D Engineer, *Collins Aerospace*, Blagnac, France — Sept 2023–Present

- Develop **multi-sensor estimation algorithms** for aerospace autonomous systems in collaboration with ISAE-SUPAERO.
- Develop **camera-radar sensor fusion** methods for **anti-collision systems** in the context of **ACAS X**-related applications.
- Design robust methods for **state estimation**, **camera pose estimation**, and **2D–3D registration** under uncertainty.
- Build pipelines combining **camera**, **radar**, **IMU**, and **GNSS** data, with strong emphasis on algorithmic robustness and integration constraints.
- Design **synthetic datasets** and conduct **real-world drone acquisition campaigns** to validate algorithms in realistic operating conditions.
- Contribute to international research publications and collaborate with multidisciplinary teams in an international environment.

R&D / Deep Learning Engineer Intern, *Thales LAS*, Paris, France — Mar–Sept 2023

- Developed deep learning methods for image restoration on near-infrared and infrared datasets.
- Benchmarked, fine-tuned, and evaluated models across several experimental settings.
- Supported technical decision-making through comparative analysis of robustness and performance.

R&D Engineer Intern, *TéSA Laboratory*, Toulouse, France — June–Sept 2022

- Developed a **Bayesian geometric modeling** method for partial and noisy LiDAR point clouds with robust outlier handling.
- Contributed to model design, validation on synthetic data, and scientific communication, resulting in an international publication.

Education

Doctoral Degree, *ISAE-SUPAERO*, Toulouse, France — 2024–Present

PhD on **AI methods for perception and estimation under uncertainty**, in partnership with Collins Aerospace.

Engineering Degree, *ENSEEIHT*, Toulouse, France — 2020–2023

Specialization in **signal processing**, **machine learning**, **computer vision**, and **applied mathematics**.

Hard Skills

Programming: Python, C++, MATLAB, Linux, Git

GNC / Estimation: Sensor fusion, state estimation, Kalman filtering, filtering, probabilistic modeling, nonlinear optimization

Sensors / Systems: IMU, GNSS, camera, radar, LiDAR, real-world experimentation

AI / Vision: Computer vision, machine learning, deep learning, model evaluation

Soft Skills

Communication
Problem-solving
Teamwork
Rigour

Languages

French (Native)
English (C1)

Interests

Robotics projects
Academic tutoring
Running (Toulouse Half Marathon)